

When Does Learning Help Trellis Detection? Matched Baselines, Structure, and Complexity of Model-Based Deep Viterbi Receivers

Gil Zukerman

Abstract—Model-based deep receivers such as ViterbiNet keep the Viterbi trellis and learn the channel-dependent branch metric from pilots, adapting it online on decoded words. They are usually evaluated against a Viterbi detector with perfect or deliberately corrupted channel state information (CSI), neither of which receives the information the learned receiver uses. We re-evaluate this design space with matched-information baselines, paired Monte Carlo comparisons, controlled ablations, and a short analysis. On a coded BPSK intersymbol-interference channel with COST 2100 taps, a Viterbi detector that estimates its four taps by least squares (LS) from the same pilots and the same code-gated decisions has the lowest point-estimate SER of all no-CSI receivers from 0 to 14 dB, is significantly better than the best learned receiver at most SNRs from 6 to 12 dB on shared draws, and is never significantly worse; it stays within 3–30% of the perfect-CSI bound, in line with the 0.13 dB estimation loss we predict for LS from 136 samples. These conclusions hold when the reference implementation’s oracle adaptation gate is replaced by an implementable one. ViterbiNet’s default of 200 online steps per word overfits; 5 steps give a lower SER at every SNR and are 40 times cheaper. A 32-parameter affine metric, which we show contains the maximum-likelihood metric, matches the 7,002-parameter network; Transformer and Vision-Transformer metrics of the same size are worse at every training budget and 5–7 times costlier to adapt. Only LS-Viterbi and ViterbiNet with 5 steps come close to a real-time budget on one CPU thread. For complex QPSK the free per-branch metric needs M^L classes and fails with a realistic pilot, while tying the classes to the channel taps restores classical performance. Under fast fading, the learned receivers’ edge over a last-word LS estimate comes from memory of the channel statistics, which exponentially weighted LS reproduces; no receiver without CSI tracks the channel within a word. We release all code and data and derive design guidance for when learning can help.

Index Terms—Deep learning, model-based deep learning, symbol detection, Viterbi algorithm, channel estimation, online learning, Transformers, intersymbol interference, fast fading.

I. INTRODUCTION

DEEP learning has been applied to nearly every block of the physical layer [1], including channel estimation and detection in OFDM [2], MIMO detection [3], and sequence detection for channels without tractable models [4]. Model-based deep learning takes a middle path: it keeps the structure of a classical algorithm and learns only its model-dependent parts [5]. ViterbiNet [6] applies this idea to channels with memory. The Viterbi recursion [7], [8] is kept, and a small neural network maps each received sample to the likelihoods

of the channel states. The network is trained on pilots and adapted online on words accepted by an error-correcting code, so the receiver follows a slowly varying channel without new pilots. Meta-learning was later added to accelerate that adaptation [9], [10].

The evidence for this family rests on two comparisons: the learned receiver approaches the Viterbi algorithm with perfect CSI, and it outperforms the Viterbi algorithm given CSI with a fixed amount of uncertainty [6]. Neither baseline receives the same information as the learned receiver. The first is given the channel; the second is given a corrupted channel and no way to correct it. A classical receiver that estimates the channel from the same pilots and the same accepted decisions, the adaptive maximum-likelihood sequence estimator [11], [12], is the comparison that isolates what learning contributes. It is absent from the evaluations of learned trellis detectors that we are aware of.

This paper re-examines the design space with matched information, paired Monte Carlo comparisons where draws are shared, and ablations of each design choice. Our contributions are:

- 1) *A matched-information baseline.* An LS Viterbi receiver that uses the same pilot and code-gated decisions as ViterbiNet has the lowest point-estimate SER of all no-CSI receivers from 0 to 14 dB, is never significantly worse on paired draws, and tracks the perfect-CSI bound (Section VII-A).
- 2) *Analysis.* We show that the affine metric contains the maximum-likelihood branch metric, predict the SNR loss of LS estimation, and count operations per word as a function of the channel memory (Section V).
- 3) *Adaptation budget.* The default of 200 online steps per accepted word overfits; 5 steps give a lower SER at every SNR and are 40 times cheaper (Section VII-D).
- 4) *Structure over size.* A 32-parameter affine metric matches the 7,002-parameter network, and larger networks are not better (Section VII-E).
- 5) *Sequence architectures and cost.* Transformer [13] and Vision-Transformer [14] metrics of the same size are worse at every offline training budget and costlier to run and adapt; we map every receiver on an accuracy–compute plane and against a real-time budget (Sections VII-F–VII-G).
- 6) *Scaling to higher-order modulation.* For QPSK the per-branch metric needs M^L classes and fails with a realistic pilot; tying the classes to the channel taps restores clas-

sical performance, while symbol-by-symbol classifiers need an equalizer structure to track (Section VII-H).

- 7) *Fast fading*. The learned receivers' advantage over a last-word LS estimate comes from memory of the channel statistics, which exponentially weighted LS reproduces over a wide range of forgetting factors; no receiver without CSI tracks the channel within a word (Section VII-I).

All code, raw results, and figure scripts are released so that every number can be regenerated.

II. RELATED WORK

Learned detectors. Early data-driven detectors replaced the receiver with a generic network [1], [2] or unfolded an iterative algorithm into layers [3]. Sliding bidirectional recurrent networks detect sequences without a channel model [4]. Model-based designs such as ViterbiNet keep an inference algorithm and learn a factor of it [5], [6]; online meta-learning reduces the data and iterations needed to adapt [9], [10].

Classical adaptive sequence estimation. Adaptive MLSE with a decision-directed channel estimator dates to Magee and Proakis [11]. Per-survivor processing (PSP) embeds the estimator in the trellis so that each survivor path carries its own channel estimate [12]. Least-mean-squares and recursive least-squares estimators, including exponential forgetting, are standard tools for tracking [15]. These methods use exactly the side information ViterbiNet uses: pilots and past decisions.

Sequence models. Transformers [13] and their patch-based variants [14] have been proposed for many receiver tasks. We test whether attention over a window of samples improves the branch metric when it is embedded in the same trellis and trained under the same budget.

III. SYSTEM MODEL

A. BPSK channel with memory

Information words of 120 bits are encoded with a Reed–Solomon (RS) code over $\text{GF}(2^8)$ with two parity symbols [16], giving 136 coded bits, and BPSK-modulated. The received sequence is

$$y_t = \sum_{k=0}^{L-1} h_k^{(w)} s_{t+L-1-k} + n_t, \quad n_t \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

with memory $L = 4$, $\text{SNR} = 1/\sigma^2$, and state $x_t = (s_t, \dots, s_{t+L-1}) \in \{\pm 1\}^4$ (16 states). The taps of word w are the magnitudes of a fixed COST 2100 channel trace [17] of 300 blocks (Fig. 1a). Within each block of 50 words the taps decay smoothly; between blocks they jump, so a frame sees two abrupt changes that the receivers must follow from decisions alone. A frame has 125 words; every 25th word is a pilot, so each frame carries 5 pilots and 120 data words.

B. Fast-fading extension

To test tracking within a word, the taps of Section VII-I vary sample by sample:

$$h_k(t) = \bar{h}_k^{(w)} \left(\sqrt{\frac{\kappa}{\kappa+1}} + \sqrt{\frac{1}{\kappa+1}} g_k(t) \right), \quad (2)$$

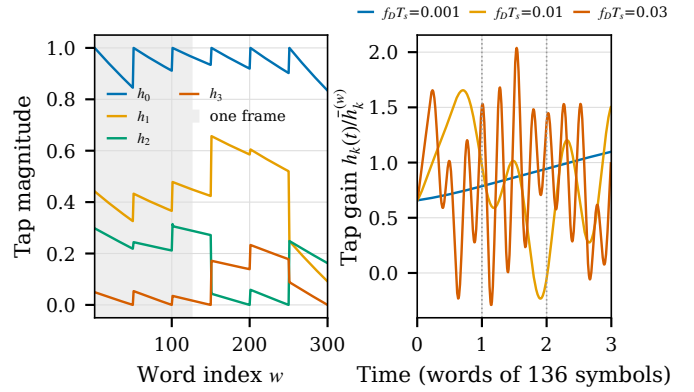


Fig. 1. Channels. (a) COST 2100 tap magnitudes per word; the shaded region is the 125-word frame used for evaluation. (b) One realization of the fast-fading tap gain over three words for three Doppler values (dotted lines: word boundaries).

TABLE I
RECEIVERS COMPARED (BPSK, $L = 4$, $T = 136$ SAMPLES PER WORD).

Receiver	Branch metric	Param.	Side information	Update per accepted word
Perfect CSI	Gaussian, true $\mathbf{h}$	–	true taps	none
Noisy CSI	Gaussian, $\mathbf{h} + \epsilon$	–	corrupted taps	none
LS-Viterbi	Gaussian, $\hat{\mathbf{h}}$	4	pilots, accepted words	one LS solve
RLS	Gaussian, $\hat{\mathbf{h}}$	4	pilots, accepted words	weighted LS
PSP-LMS	Gaussian, per survivor	4×16	pilots, own survivors	LMS per symbol
ViterbiNet	MLP $1 \rightarrow 100 \rightarrow 58 \rightarrow 16$	7,002	pilots, accepted words	K Adam steps
VNet-affine	$\mathbf{a}y_t + \mathbf{b}$	32	pilots, accepted words	K Adam steps
Transformer V2–V4, ViT	attention, 4-sample window	$\approx 7,000$	pilots, accepted words	K Adam steps

where $\bar{h}^{(w)}$ is the COST 2100 trace, $\kappa = 3$ is a Rician factor, and $g_k(t)$ are independent real, unit-variance Jakes processes (sums of 16 sinusoids) [18] with normalized Doppler $f_D T_s \in \{0.001, 0.005, 0.01, 0.03\}$, continuous across word boundaries. Fig. 1b shows the resulting tap gain: at $f_D T_s = 0.001$ it drifts slowly over three words, at 0.01 it changes appreciably over about 40 symbols, and at 0.03 it changes sign several times within one word.

C. Complex QPSK channel

For Section VII-H, QPSK symbols pass through $L = 3$ complex Rayleigh taps with an exponential power-delay profile (power ratio 0.5 per tap, unit total power). The taps evolve from word to word as a first-order autoregressive process with correlation 0.99, or, in the fast-fading variant, as complex Jakes processes. Words carry 120 symbols between known start and end symbols; a frame has one pilot word and 24 data words. The trellis has $M^{L-1} = 16$ states and $M^L = 64$ branches.

IV. RECEIVERS

All receivers share the structure of Fig. 2: one Viterbi recursion over the 16-state trellis, an RS decoder, and a gate that decides which decoded words are used for adaptation (Section VI). They differ only in the branch metric $\ell_t(x)$ and in the rule that updates it from pilots and accepted words. Table I summarizes them.

A. Classical receivers

Perfect CSI: $\ell_t(x) = -(y_t - \mathbf{h}^T \mathbf{s}(x))^2/2$ with the true taps (the bound).

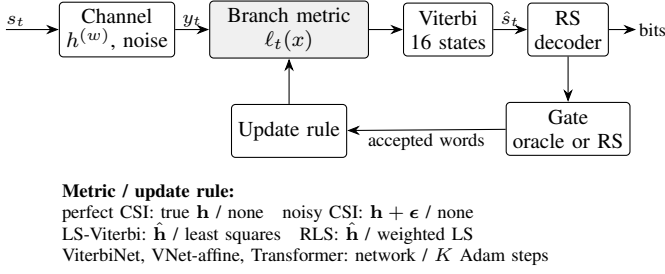


Fig. 2. Common receiver structure. Only the shaded branch-metric block and its update rule differ between receivers; every adaptive receiver adapts on the same pilots and accepted words.

Noisy CSI: the same metric with taps $1, \dots, L-1$ perturbed by Gaussian noise of standard deviation ρ times the RMS tap magnitude, $\rho \in \{0.25, 0.5, 0.75, 1\}$. This is the CSI-uncertainty baseline of [6].

LS-Viterbi (matched information): the metric uses $\hat{\mathbf{h}} = \arg \min_{\mathbf{h}} \sum_t (y_t - \mathbf{h}^T \mathbf{s}(x_t))^2$, recomputed on each pilot and on each data word accepted by the gate, with that word's labels. These are the words and labels the learned receivers adapt on.

LS with forgetting (RLS): $\hat{\mathbf{h}} = \mathbf{R}^{-1} \mathbf{r}$ with $\mathbf{R} \leftarrow \lambda \mathbf{R} + \mathbf{X}^T \mathbf{X}$ and $\mathbf{r} \leftarrow \lambda \mathbf{r} + \mathbf{X}^T \mathbf{y}$ over accepted words ($\lambda = 0.8$ unless stated), i.e., exponentially weighted LS [15] with an effective memory of $1/(1 - \lambda)$ words.

PSP-LMS: each survivor carries its own tap estimate, updated by least mean squares along its own path at every trellis step [12] (step size $\mu = 0.01$ unless stated).

B. Learned receivers

ViterbiNet: a fully connected network $1 \rightarrow 100 \rightarrow 58 \rightarrow 16$ (sigmoid, then ReLU; 7,002 parameters) maps y_t to 16 state scores, trained with cross-entropy against the true state [6]. Offline training uses 25 minibatches of 300 words with Adam [19] (learning rate 10^{-3}); online, K Adam steps are taken on each pilot and each accepted data word ($K = 200$ by default).

VNet topologies: the same pipeline with hidden widths from 4 to 64, two-layer variants, and no hidden layer. The last, VNet-affine, computes $\mathbf{a}y_t + \mathbf{b} \in \mathbb{R}^{16}$ (32 parameters).

Transformer metrics (all about 7,000 parameters, window of 4 samples, model width 16): the original encoder with 8 heads; V2 with 2 heads (head width 8) and sinusoidal positions; V3, which replaces the linear embedding with an MLP; V4, an MLP feature extractor, one attention layer, and an MLP head; and ViT variants that treat non-overlapping or overlapping 4-sample patches as tokens [13], [14]. All output 16 state scores for the same trellis.

C. QPSK receivers

Free per-branch metrics extend ViterbiNet directly: an affine map (192 parameters) or MLP (9,934 parameters) from $(\Re y_t, \Im y_t)$ to the 64 branch scores. *Tied metrics* constrain the 64 branch means to $\mu_c = \sum_k h_k s_k(c)$ with L learnable

complex taps and a learnable scale (7 parameters), optionally adding a shared residual MLP applied to $y_t - \mu_c$ (88 parameters, initialized to zero). *Symbol classifiers* skip the trellis and map a window of received samples to 4 classes: a free linear map (44 parameters), an MLP (948), or an equalizer-structured classifier $z = \mathbf{w}^H \mathbf{y} + b$ with scores $a \Re(c_m^* z)$ (21 parameters, 9-sample window). Classical references are LS-Viterbi and a linear equalizer with MMSE weights from the true taps or LS weights re-fitted on its own decisions.

V. ANALYSIS

A. The affine metric contains the ML metric

Proposition 1. *For channel (1) with known taps and equiprobable states, the posterior logits of the state given one sample are affine in y_t : $\log p(x | y_t) = (\mu_x / \sigma^2) y_t - \mu_x^2 / (2\sigma^2) + c(y_t)$ with $\mu_x = \mathbf{h}^T \mathbf{s}(x)$. Hence (i) the Bayes-optimal cross-entropy classifier of x_t from y_t lies in the affine family, and (ii) using its logits as branch metrics yields exactly the maximum-likelihood Viterbi decisions.*

Proof. $\log p(y_t | x) = -(y_t - \mu_x)^2 / (2\sigma^2) + \text{const}$ and $\log p(x)$ is constant, so Bayes' rule gives the stated form with $c(y_t) = -y_t^2 / (2\sigma^2) - \log \sum_{x'} p(y_t | x') + \text{const}$. The minimizer of the expected cross-entropy is the true posterior, which is affine up to $c(y_t)$; since softmax is invariant to the state-independent $c(y_t)$, an affine layer attains it. For (ii), $c(y_t)$ adds the same amount to every branch at time t , so it changes no comparison in the add-compare-select recursion, and the remaining terms are the Gaussian branch metric up to scale. $\square$

Proposition 1 explains why VNet-affine can match a much larger network on this channel: the extra capacity has nothing to represent. It also shows how over-parameterized the affine metric still is: its 32 parameters describe 16 means μ_x that are fixed by 4 taps. It does not imply that 32 parameters suffice when the noise is non-Gaussian or the channel is nonlinear.

B. Estimation loss of LS-Viterbi

Let $\hat{\mathbf{h}}$ be the LS estimate from one word of T samples with correct labels. Its error has covariance $\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$, so the residual $y_t - \hat{\mathbf{h}}^T \mathbf{s}(x)$ seen by the metric of state x has variance $\sigma^2 (1 + \mathbf{s}(x)^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{s}(x))$. If the shifted-symbol columns of $\mathbf{X}$ are nearly orthogonal, $\mathbf{X}^T \mathbf{X} \approx T \mathbf{I}_L$ and the excess is L/T . This is an assumption about the training words, not a general LS result, so we checked it for our RS-coded words: averaged over 2,000 random words and the 16 states, $\mathbf{s}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{s} = 0.0301$ against $L/T = 0.0294$, and the median condition number of $\mathbf{X}^T \mathbf{X} / T$ is 1.39. To first order LS-Viterbi therefore behaves like the perfect-CSI receiver at an SNR lower by

$$\Delta \approx 10 \log_{10} \left(1 + \frac{L}{T} \right) \text{ dB} = 0.126 \text{ dB} \quad (L = 4, T = 136), \quad (3)$$

or 0.129 dB with the exact average for the coded words. Shifting the measured perfect-CSI curve by Δ predicts SER ratios of 1.03, 1.07, 1.16, and 1.27 at 4, 7, 10, and 12 dB; we measure 1.04, 1.06, 1.13, and 1.30 (Fig. 5). At 13–14 dB

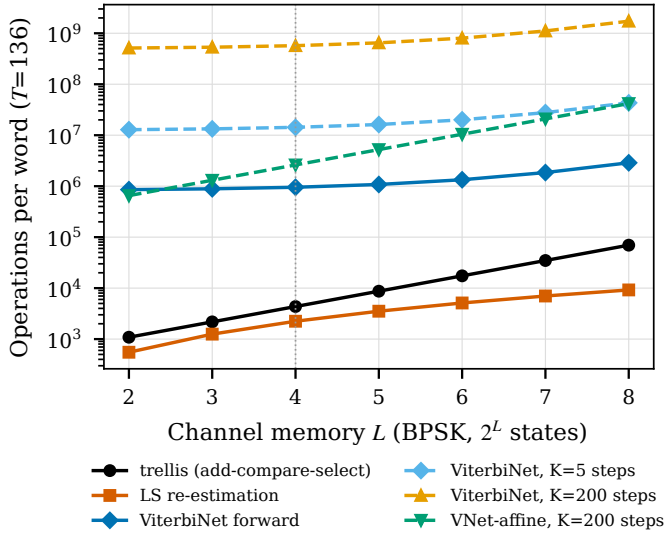


Fig. 3. Operations per 136-sample word vs. channel memory (BPSK). Network costs scale the $1 \rightarrow 100 \rightarrow 58 \rightarrow 2^L$ ViterbiNet and the $2 \cdot 2^L$ -parameter affine metric; adaptation counts forward and backward passes. Dotted line: $L = 4$ used in this paper.

the prediction (1.31–1.49) slightly exceeds the measurement (1.22–1.25); there the 14 dB point rests on 4 LS error events, and the prediction interpolates a perfect-CSI curve that falls by an order of magnitude per dB. The prediction ignores label errors in accepted words and the jumps of Fig. 1a; that it fits this closely suggests that both have little effect and that the remaining gap to the bound is the estimation noise of four unknowns.

C. Complexity

For a word of T samples, the trellis costs $O(T2^{L+1})$ add-compare-select operations for every receiver. A per-sample metric with P parameters adds about TP multiply-accumulates for the forward pass and $3KTP$ for K adaptation steps (forward and backward), while LS re-estimation costs $O(TL^2 + L^3)$, i.e., about 2,200 multiply-accumulates for $T = 136$ and $L = 4$, fewer than the 7,002-parameter network spends on a single sample. Fig. 3 extends these counts to other memories: the ViterbiNet output layer grows as 2^L like the trellis, so learned adaptation stays three to five orders of magnitude above the trellis, while LS re-estimation stays below the trellis for every L . Operation counts are the platform-independent measure; measured latency (Section VII-G) is instead dominated by the trellis, because our trellis is a sequential loop of 136 interpreted steps while the network runs as vectorized kernels.

VI. EVALUATION PROTOCOL

Matched information and the adaptation gate. Every receiver sees the same pilot words (one in every 25 words), and every adaptive receiver uses the same gate. We follow the reference implementation of ViterbiNet: a data word is used for adaptation if the SER of its RS-decoded bits is at most 0.02, with the RS re-encoded word as labels when that SER

is zero and the detector’s hard decisions otherwise. Computing that SER requires the transmitted bits, so this gate is an oracle, not a receiver function; it is applied identically to every adaptive receiver. We also evaluate an implementable gate that uses no ground truth: accept a word when the re-encoded decoded word differs from the hard decisions in at most one RS symbol (bounded-distance decoding with two parity symbols) and adapt on the re-encoded word (Section VII-B).

Monte Carlo. A repetition draws fresh bits and noise for one frame; the SER is computed on the RS-decoded information bits of the 120 data words (14,400 bits per repetition). Learned receivers use at least 20 repetitions (50 for the SNR sweeps); the perfect-CSI and LS receivers use at least 100 (up to 10^4 and 10^3), with the count set from a predicted SER to target about 100 errors. Results are reported from 0 to 14 dB; figures extend to 15 dB, where zero-error points are drawn as the one-sided 95% bound $3/N$ for N evaluated bits. At 14 dB the SER of ViterbiNet and VNet-affine rests on 4–9 error events and that of LS-Viterbi on 4 (95% Poisson interval for 4 events: 0.3–2.6 times the estimate), so ratios there are indicative only.

Shared draws and paired comparisons. The perfect-CSI, LS-Viterbi, ViterbiNet ($K = 5$), and VNet-affine receivers read repetition $r < 200$ from a shared cache of generated (bits, noise) draws, so their first repetitions are paired; their per-repetition SERs are positively correlated (median $\rho \approx 0.5$), as shared draws imply. We report paired differences over the shared repetitions. The $K = 200$ ViterbiNet and Transformer sweeps ran with the same code on separate GPU machines with independent draws, and the noisy-CSI sweeps used a separate cache; comparisons against them are unpaired.

Provenance. All reported points were generated after we corrected a data-cache defect that had replayed one draw across all repetitions of a point; results collected before the fix were discarded. Every reported point shows repetition-to-repetition variance (standard deviation at least 1.9% of the mean), which the defect would have made zero. Architecture and topology studies at 7 dB use 20 repetitions each and report 95% confidence intervals. In the fast-fading study, evaluation data are seeded per repetition, so all receivers see identical transmissions and differences are paired. QPSK points average 64 frames of 24 data words (184,320 symbols).

Training data. Offline training of the learned receivers uses words drawn from the same COST 2100 tap sequence used for evaluation. This favors the learned receivers and makes the comparison conservative.

Hardware. Latency is measured on a single CPU thread as the median of repeated runs on one 136-sample word. The $K = 200$ ViterbiNet and Transformer SNR sweeps were run with the same code on separate GPU machines.

VII. RESULTS

A. BPSK: matched-information baseline

Fig. 4 and Table II report SER versus SNR. In point estimate, LS-Viterbi has the lowest SER of all no-CSI receivers at every SNR from 0 to 14 dB, and it stays within 3–30% of the perfect-CSI SER: $2.4e-2$ versus $2.2e-2$ at 7 dB, $5.7e-5$ versus $4.3e-5$ at 12 dB, and $2.8e-7$ versus $2.2e-7$ at 14 dB.

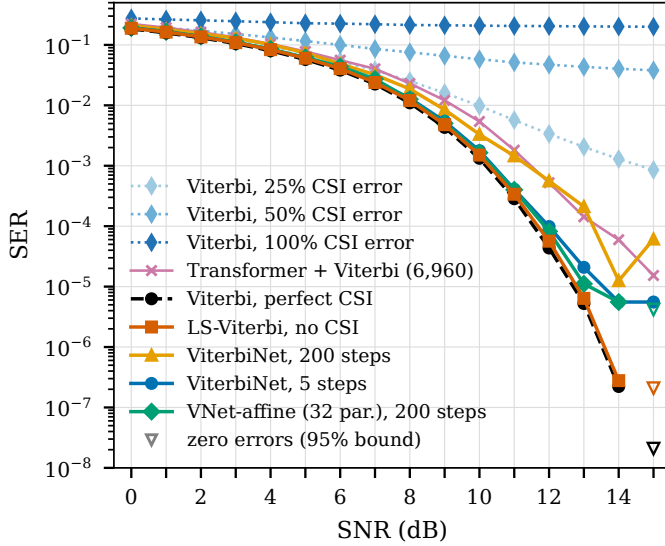


Fig. 4. BPSK SER vs. SNR. LS-Viterbi receives the same pilots and accepted decisions as the learned receivers; noisy-CSI curves perturb the taps by a fixed fraction. Hollow triangles: no errors observed (95% upper bound).

TABLE II
BPSK SER OF THE RS-DECODED BITS.

Receiver	Param.	0 dB	7 dB	10 dB	12 dB	14 dB
Viterbi, perfect CSI	–	0.184	2.2e-2	1.3e-3	4.3e-5	2.2e-7
Viterbi, 25% CSI error	–	0.192	3.8e-2	9.8e-3	3.4e-3	1.3e-3
LS-Viterbi (no CSI)	4	0.189	2.4e-2	1.5e-3	5.7e-5	2.8e-7
ViterbiNet, $K=200$	7,002	0.210	3.2e-2	3.3e-3	5.6e-4	1.2e-5
ViterbiNet, $K=5$	7,002	0.191	2.5e-2	1.8e-3	9.9e-5	5.6e-6
VNet-affine, $K=200$	32	0.191	2.7e-2	1.6e-3	8.2e-5	5.6e-6
Transformer, $K=200$	6,960	0.222	4.0e-2	5.4e-3	5.2e-4	6.0e-5

Relative to ViterbiNet with its default adaptation, LS-Viterbi has 1.3 times lower SER at 7 dB, 9.8 times at 12 dB, and 45 times at 14 dB. With $K = 5$ the ratios are 1.07, 1.7, and 20. On the shared draws (Table III), LS-Viterbi is significantly better than ViterbiNet with $K = 5$ at 1, 6, 7, and 9–12 dB and than VNet-affine at 0, 2–7, and 11 dB, and never significantly worse; at 13–14 dB the paired differences are within noise, so the high-SNR ranking is a point-estimate ranking only.

Fig. 5 divides every curve by the perfect-CSI SER. Up to 6 dB every receiver is within 1.5 times the bound, and the choice of receiver matters little. The curves separate as the SNR grows: the gap of LS-Viterbi stays flat at 1.0–1.3, as (3) predicts, whereas ViterbiNet with $K = 5$ and VNet-affine grow to 1.9–4 times at 12–13 dB and about 25 times at 14 dB, the default ViterbiNet to 13–56 times at 12–14 dB, and the Transformer to 270 times at 14 dB. A learned metric fitted by stochastic gradient steps carries an error floor that is invisible at low SNR and dominant at high SNR; one LS solve does not.

B. An implementable adaptation gate

The results above use the oracle gate of the reference implementation (Section VI). Table IV repeats four adaptive receivers with the implementable RS gate, starting from the same offline weights and evaluating the same repetitions, so

TABLE III
PAIRED SER DIFFERENCES, LS-VITERBI MINUS LEARNED RECEIVER, OVER THE MONTE CARLO REPETITIONS BOTH EVALUATED ON THE SAME DRAWS (MEAN $\pm$ 95% CI; *: CI EXCLUDES ZERO; NEGATIVE: LS-VITERBI BETTER; -: PER-REPETITION VALUES NOT SAVED FOR THAT POINT).

SNR (dB)	vs. ViterbiNet, $K=5$	vs. VNet-affine	Reps
0	$-1.6 \times 10^{-3} \pm 1.9 \times 10^{-3}$	$-2.4 \times 10^{-3} \pm 1.9 \times 10^{-3}$ *	20
1	$-2.6 \times 10^{-3} \pm 2.2 \times 10^{-3}$ *	$-1.4 \times 10^{-3} \pm 1.7 \times 10^{-3}$	20
2	$-7.5 \times 10^{-4} \pm 1.4 \times 10^{-3}$	$-1.9 \times 10^{-3} \pm 1.5 \times 10^{-3}$ *	20
3	$-1.4 \times 10^{-3} \pm 1.5 \times 10^{-3}$	$-1.9 \times 10^{-3} \pm 1.7 \times 10^{-3}$ *	20
4	$-7.6 \times 10^{-4} \pm 1.2 \times 10^{-3}$	$-2.2 \times 10^{-3} \pm 1.4 \times 10^{-3}$ *	20
5	–	$-2.6 \times 10^{-3} \pm 1.4 \times 10^{-3}$ *	20
6	$-1.8 \times 10^{-3} \pm 1.0 \times 10^{-3}$ *	$-3.8 \times 10^{-3} \pm 2.1 \times 10^{-3}$ *	20
7	$-1.9 \times 10^{-3} \pm 1.0 \times 10^{-3}$ *	$-3.6 \times 10^{-3} \pm 3.0 \times 10^{-3}$ *	20
8	–	$-4.6 \times 10^{-4} \pm 1.2 \times 10^{-3}$	20
9	$-7.8 \times 10^{-4} \pm 4.6 \times 10^{-4}$ *	$-3.9 \times 10^{-4} \pm 5.5 \times 10^{-4}$	21
10	$-2.5 \times 10^{-4} \pm 1.2 \times 10^{-4}$ *	$-1.2 \times 10^{-4} \pm 1.5 \times 10^{-4}$	50
11	$-9.3 \times 10^{-5} \pm 6.6 \times 10^{-5}$ *	$-8.5 \times 10^{-5} \pm 7.2 \times 10^{-5}$ *	50
12	$-5.0 \times 10^{-5} \pm 3.7 \times 10^{-5}$ *	$-3.3 \times 10^{-5} \pm 3.5 \times 10^{-5}$	50
13	$-2.8 \times 10^{-6} \pm 1.6 \times 10^{-5}$	$+6.9 \times 10^{-6} \pm 1.1 \times 10^{-5}$	50
14	$-2.8 \times 10^{-6} \pm 1.2 \times 10^{-5}$	$-2.8 \times 10^{-6} \pm 1.2 \times 10^{-5}$	50

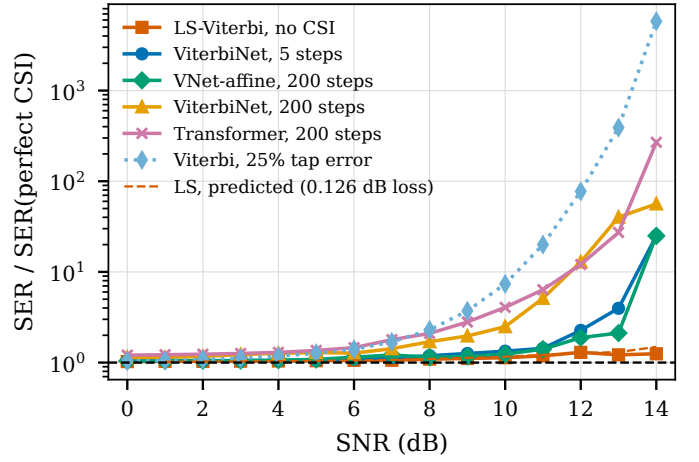


Fig. 5. SER relative to the perfect-CSI Viterbi detector, 0–14 dB. Dashed red: LS-Viterbi predicted from (3). At 14 dB the ViterbiNet and VNet-affine points rest on 4–9 error events.

each receiver's two rows are paired. The implementable gate is stricter at low SNR: at 7 dB it accepts 21–33% of data words against 30–44% for the oracle gate. It is also cleaner, because it always adapts on a codeword: 15–22% of the accepted words carry any label error, against 37–46% for the oracle gate, whose labels are raw hard decisions whenever the decoded SER is nonzero. At 10 and 12 dB both gates accept most words (83–100%), and both label errors and differences shrink. The two effects cancel: no receiver's SER changes by more than the 95% confidence interval of either run, and LS-Viterbi keeps the lowest SER under both gates. The conclusions of this section therefore do not depend on the oracle gate.

C. The noisy-CSI baseline

Fig. 6 explains the original comparison. With 25% tap error the Viterbi detector improves only slowly with SNR and reaches 1.3×10^{-3} at 14 dB, 5,800 times the bound (Fig. 5); at 50–100% error it is flat at 0.04–0.2. ViterbiNet beats every

TABLE IV
ORACLE VS. IMPLEMENTABLE (RS) ADAPTATION GATE ON THE SAME
DRAWS: SER, FRACTION OF DATA WORDS ACCEPTED, AND FRACTION OF
ACCEPTED WORDS WITH WRONG LABELS.

SNR	Receiver	SER		Accepted		Wrong labels	
		oracle	RS	oracle	RS	oracle	RS
7	LS-Viterbi	2.4×10^{-2}	2.4×10^{-2}	0.44	0.33	0.37	0.149
	ViterbiNet $K=5$	2.5×10^{-2}	2.5×10^{-2}	0.42	0.30	0.38	0.165
	VNet-affine $K=200$	2.5×10^{-2}	2.6×10^{-2}	0.43	0.29	0.38	0.164
	ViterbiNet $K=200$	3.2×10^{-2}	3.2×10^{-2}	0.30	0.21	0.46	0.219
10	LS-Viterbi	1.5×10^{-3}	1.6×10^{-3}	0.97	0.93	0.04	0.003
	ViterbiNet $K=5$	1.8×10^{-3}	1.7×10^{-3}	0.96	0.93	0.04	0.004
	VNet-affine $K=200$	1.7×10^{-3}	1.7×10^{-3}	0.97	0.93	0.05	0.005
	ViterbiNet $K=200$	4.1×10^{-3}	4.4×10^{-3}	0.91	0.83	0.09	0.010
12	LS-Viterbi	4.6×10^{-5}	4.6×10^{-5}	1.00	1.00	0.00	0.001
	ViterbiNet $K=5$	1.1×10^{-4}	1.0×10^{-4}	1.00	1.00	0.00	0.001
	VNet-affine $K=200$	7.9×10^{-5}	7.6×10^{-5}	1.00	1.00	0.00	0.000
	ViterbiNet $K=200$	6.7×10^{-4}	7.2×10^{-4}	0.98	0.97	0.02	0.003

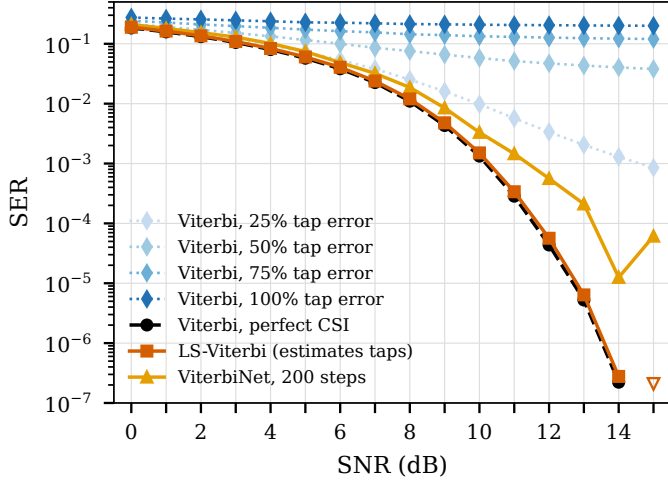


Fig. 6. The CSI-uncertainty baseline: Viterbi with taps perturbed by 25–100% of their RMS magnitude, against the perfect-CSI and LS-Viterbi receivers and ViterbiNet.

one of these curves from 6 dB on, which is the comparison reported in [6]. Nothing in the noisy-CSI receiver corrects its model. ViterbiNet corrects its model from data, and so does LS-Viterbi, with four unknowns and one least-squares solve per accepted word; on that footing the learned receiver’s advantage disappears.

D. Online adaptation budget

Fig. 7 sweeps the number of online steps K at 7 dB. For the 7,002-parameter network the SER is lowest at $K = 5$ (0.0249) and rises monotonically to 0.0315 at $K = 200$. Without adaptation it is 0.0340, so a few steps help and many steps overfit: each step fits the same 136-sample word, and after a handful of steps the network starts fitting that word’s noise. Over the whole SNR range, $K = 5$ has a lower point-estimate SER than $K = 200$ at every SNR (Table II, Fig. 5; unpaired, since the $K = 200$ sweep used independent draws) while using 40 times less adaptation compute.

The affine metric responds oppositely: it needs 50–200 steps to move from the generic offline solution to the frame’s

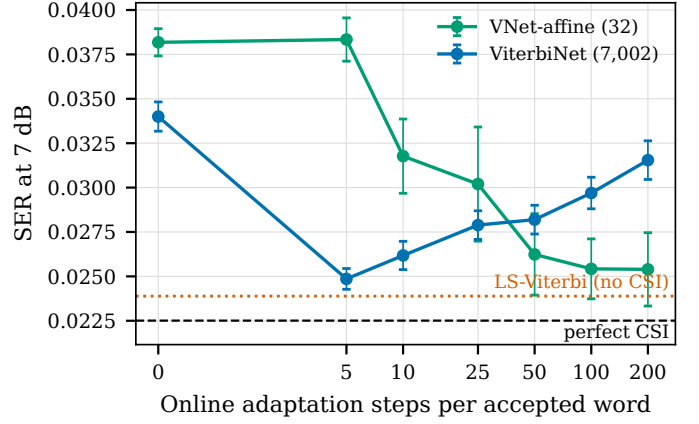


Fig. 7. SER at 7 dB vs. online steps per accepted word (25 offline minibatches; bars: 95% confidence intervals over 20 repetitions).

channel and reaches 0.0254 at $K = 100$ –200, statistically tied with the tuned network. A model with 32 parameters has too little capacity to overfit one word, but also moves more slowly per step.

E. Topology: structure over size

Fig. 8(a) shows SER against parameter count at the default budget. SER increases with size, from 0.0254 (affine) to 0.0275–0.0290 (4 to 64 hidden units), 0.0299–0.0307 (two hidden layers) and 0.0315 (7,002 parameters). Only the confidence intervals of the smallest networks overlap the affine one. Fig. 8(b) varies the offline budget: from 2 to 25 minibatches the large network stays near 0.031–0.033, whereas the affine metric improves with more offline training. More capacity does not buy accuracy, because the metric the network must represent is quadratic in y_t with four unknowns (Proposition 1).

F. Sequence architectures and training budget

Fig. 9 compares metrics of about 7,000 parameters at 7 dB under a training-budget search: starting at 25 offline minibatches, the budget grows by 25 while the SER improves and is then refined by a binary search around the best value, with fixed-budget runs up to 1,000 minibatches. The MLP is best at every budget (0.031–0.033). The best attention-based metric, a ViT with overlapping patches, reaches 0.033 after 250 minibatches; Transformer V2, V3, and V4 stay between 0.037 and 0.041; the original 8-head encoder reaches 0.040; and a ViT with non-overlapping patches fails (0.107), because a 4-sample patch splits the channel memory across tokens. Attention over a 4-sample window adds no information the trellis does not already use, and the extra parameters are spent on mixing that the channel structure makes unnecessary. Over the SNR sweep the Transformer is the worst adaptive receiver at every SNR (Fig. 5).

G. Latency, real-time budget, and the accuracy–compute trade-off

Fig. 10 breaks detection latency down on one CPU thread. The 136-step, 16-state trellis costs 11–12 ms per word (81–

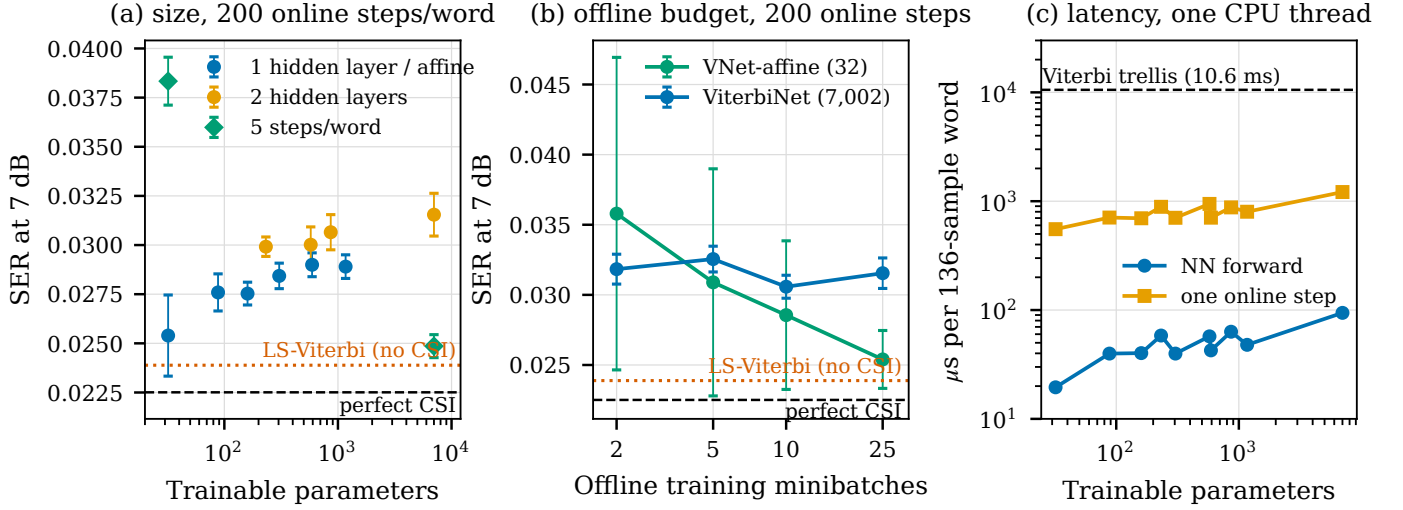


Fig. 8. ViterbiNet topology study at 7 dB. (a) SER vs. trainable parameters at 25 offline minibatches and 200 online steps (green: 5 online steps). (b) SER vs. offline minibatches. (c) Per-word latency of the network forward pass and one online step; the 16-state trellis costs about 10.6 ms per word for every topology.

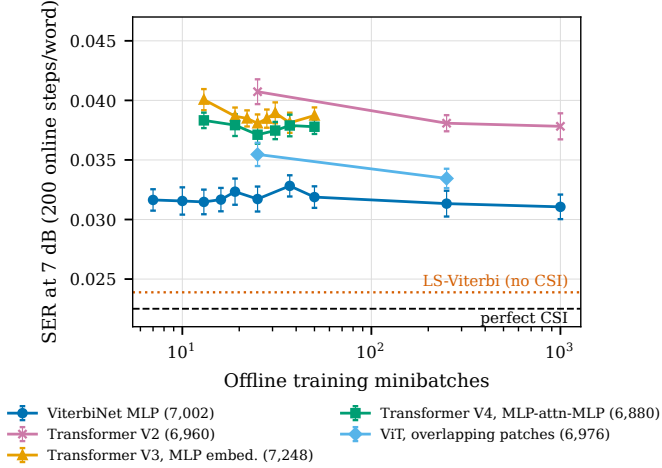


Fig. 9. SER at 7 dB vs. offline training budget for metrics of about 7,000 parameters (200 online steps; bars: 95% confidence intervals).

90 μ s per symbol) for every receiver. The network forward pass adds 0.05 ms (affine), 0.14 ms (ViterbiNet), and 1.4–2.4 ms (attention-based). The difference that matters is adaptation: one online step takes 0.53 ms (affine), 1.0 ms (ViterbiNet), and 4.9–7.3 ms (attention-based).

Fig. 11 adds the adaptation of one accepted word to the detection time and compares the total with the duration of a word at 8 ksymbol/s (0.125 ms per symbol, 17 ms per word). LS-Viterbi needs 11.4 ms and ViterbiNet with $K = 5$ 16.8 ms, both inside the budget on a single thread of an interpreted implementation. VNet-affine with $K = 200$ (117 ms), the default ViterbiNet (220 ms), and the Transformers (0.99–1.5 s) exceed it by one to two orders of magnitude and would need batching across words, dedicated hardware, or fewer steps.

Fig. 12 plots SER at 7 dB against adaptation compute per accepted word for every topology and budget. The LS receiver sits at the lower left: 0.02 ms for one solve and an SER below

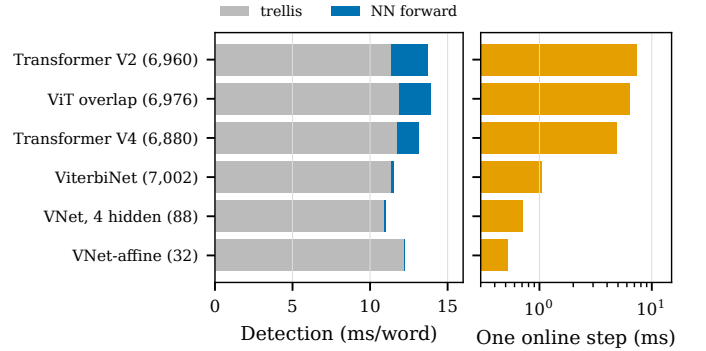


Fig. 10. Per-word latency on one CPU thread: detection (trellis plus network forward pass) and one online adaptation step.

every learned point. Among the learned receivers the Pareto front is formed by ViterbiNet with $K = 5$ (5 ms) and the affine metric with $K = 200$ (105 ms); the default ViterbiNet (209 ms) is dominated by both, and the Transformers (about 1 s) are dominated by all. A smaller network is therefore not automatically a cheaper receiver: the number of adaptation steps it needs matters as much as its size.

H. QPSK: tying the branch classes

Fig. 13 and Table V show the complex QPSK channel from 0 to 14 dB. The free per-branch metrics, the direct extension of ViterbiNet, fail: at 14 dB they reach 0.44 (affine) and 0.57 (MLP), 14 and 18 times the LS-Viterbi SER, and their gap grows with SNR. The tied metric (7 parameters) stays within 1.1–1.8 times LS-Viterbi at every SNR, and the tied metric with a residual MLP (88 parameters) within 1.2–2.9 times except at 14 dB, where it is 2.5 times better (1.3×10^{-2} against 3.2×10^{-2}). Neighboring SNR points use independent channel draws and the tracking receivers propagate decision errors, so single points fluctuate; comparisons are meaningful within an SNR point, where all receivers share the draws.

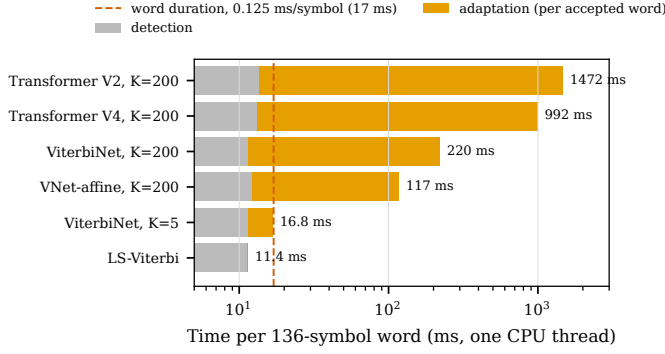


Fig. 11. Detection plus adaptation time per accepted 136-symbol word on one CPU thread, against the word duration at 0.125 ms per symbol.

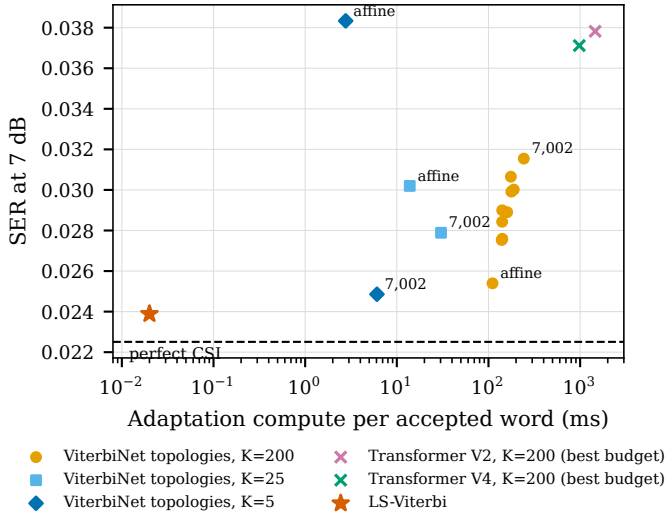


Fig. 12. Accuracy vs. adaptation compute at 7 dB. Circles, squares, diamonds: ViterbiNet topologies (4 hidden units to 7,002 parameters, and the affine metric) at $K = 200, 25$, and 5 online steps. Crosses: Transformers at their best offline budget. Star: LS-Viterbi.

Why the free metric fails: data, not tracking. Fig. 14 removes the drift (static channel, 14 dB) and varies the pilot. LS-Viterbi and the tied metric reach $4.5\text{--}8.4 \times 10^{-4}$ with a single 120-symbol pilot word and gain nothing from longer pilots. The free per-branch affine metric stays at 0.24 with one pilot word (0.13 with ten times more training steps); it needs 5 pilot words and 2,000 steps to reach 1.2×10^{-3} , within 2 times of the tied metric, at five times the pilot overhead. The free MLP never goes below 0.10, even with 10 pilot words. The failure is therefore one of data: a 120-symbol pilot visits each of the 64 branches about twice on average, and about 10 branches not at all, so a metric that learns each branch mean separately cannot be trained from one pilot. The tied metric estimates 3 taps from the same 120 symbols and generalizes to every branch, which is why it matches LS-Viterbi with a single pilot.

Why symbol classifiers need structure. A free linear or MLP 4-class classifier trained on the pilot and adapted by cross-entropy on its own decisions stays at 0.21–0.29 at 12–14 dB. Fig. 15 shows why: its per-word SER climbs from 0.16 to 0.38 across the frame at 12 dB and from 0.10 to 0.28 at 14 dB.

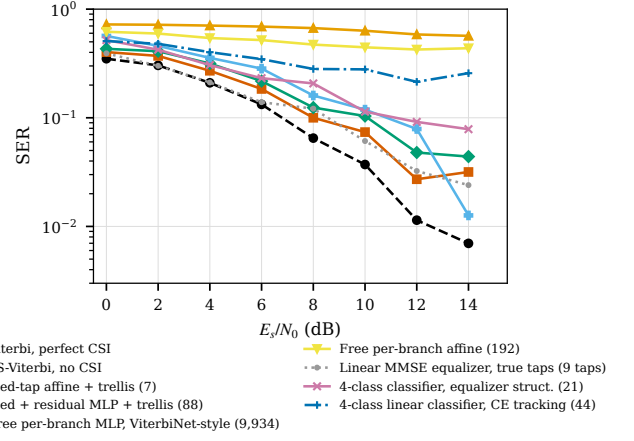


Fig. 13. QPSK, 3 complex taps drifting between words, one 120-symbol pilot per 25 words. Parentheses: trainable parameters.

TABLE V
QPSK SER (64 FRAMES $\times$ 24 DATA WORDS PER POINT).

Receiver	Param.	4 dB	8 dB	12 dB	14 dB
Viterbi, perfect CSI	–	0.21	6.5×10^{-2}	1.1×10^{-2}	7.0×10^{-3}
LS-Viterbi	–	0.27	0.10	2.7×10^{-2}	3.2×10^{-2}
Tied affine + trellis	7	0.32	0.12	4.8×10^{-2}	4.4×10^{-2}
Tied + residual MLP + trellis	88	0.36	0.16	7.9×10^{-2}	1.3×10^{-2}
Free per-branch affine	192	0.54	0.47	0.43	0.44
Free per-branch MLP	9,934	0.71	0.67	0.59	0.57
4-class linear, CE tracking	44	0.40	0.28	0.21	0.26
4-class MLP, CE tracking	948	0.41	0.30	0.24	0.29
4-class, equalizer structure	21	0.31	0.21	9.2×10^{-2}	7.9×10^{-2}
Linear equalizer, LS weights	–	0.30	0.20	8.8×10^{-2}	7.9×10^{-2}
Linear MMSE, true taps	–	0.21	0.12	3.2×10^{-2}	2.4×10^{-2}

Confident decisions produce almost no cross-entropy gradient, so self-training never follows the drifting channel, and a code gate at SER 0.02 would never open. Giving the classifier an equalizer structure and a decision-directed LMS update holds it at 0.07–0.10 across the frame and brings it to 7.9×10^{-2} at 14 dB, equal to a linear equalizer whose weights are re-fitted by LS on its own decisions (Table V; the two agree within 10% at every SNR). The trellis receivers degrade across the frame as well, LS-Viterbi from 0.012 to 0.036 and the tied metric from 0.014 to 0.056 at 14 dB, but remain below every symbol classifier.

The window limits linear equalization. Fig. 16 varies the equalizer window. With true taps the MMSE equalizer improves from 0.11 (3 samples) to 2.1×10^{-2} (9) and 1.7×10^{-2} (15) at 14 dB, still 2.4 times the perfect-CSI trellis. With LS weights fitted on its own decisions it stalls at 0.066–0.069 from 9 samples on, because decision errors bias the fit. The 9-sample window of the equalizer-structured classifier is therefore near the best a decision-directed linear receiver can do, and the remaining gap to the trellis is the cost of symbol-by-symbol detection.

I. Fast fading

Fig. 17 and Table VI show BPSK at 7 dB when the taps vary within the word. Because all receivers see the same transmissions, we report paired differences. At $f_D T_s = 0.001$ the learned receivers beat LS-Viterbi on the last accepted

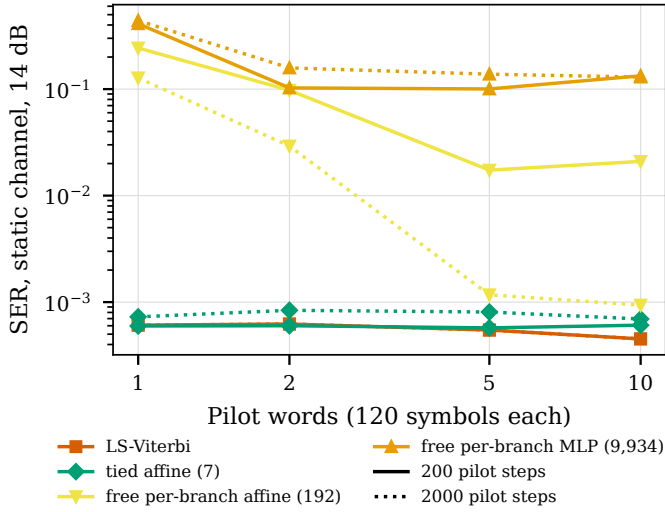


Fig. 14. QPSK on a static channel at 14 dB: SER vs. pilot length with 200 (solid) or 2,000 (dotted) pilot training steps. Parentheses: trainable parameters.

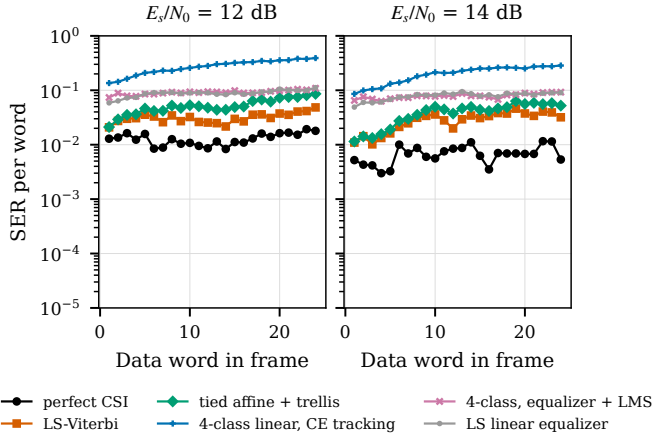


Fig. 15. QPSK per-word SER across the 24 data words of a frame (64 frames; one pilot before word 1). Words without errors are not drawn.

word by 0.039 SER and PSP-LMS by 0.017. The source of that advantage is memory rather than tracking: exponentially weighted LS with forgetting factor 0.8 ties VNet-affine up to $f_D T_s = 0.01$ and beats it at 0.03, and it is slightly better than ViterbiNet with $K = 5$ at $f_D T_s = 0.001$ and 0.03. After a deep fade, the last accepted word is a poor estimate of the next one; an estimator that averages over past words, whether by offline training on the channel statistics or by exponential forgetting, is not misled by it. No receiver without CSI tracks the channel within a word: all sit near 0.14, against 0.07 for a genie that knows the per-sample taps.

Sensitivity to the tracker settings. The classical trackers above use a single step size and forgetting factor. Fig. 18 sweeps both on 20 paired repetitions and reports the difference to VNet-affine on the same draws. PSP-LMS is sensitive to its step size. Its best setting is $\mu = 0.01$ at $f_D T_s = 0.001$ and $\mu = 0.003$ at 0.01 and 0.03; there it is within 0.002–0.005 of VNet-affine at the two higher Doppler values but 0.013±0.015 worse at 0.001, where a per-symbol LMS update is too noisy to beat averaging over words, and $\mu = 0.05$ diverges (+0.09

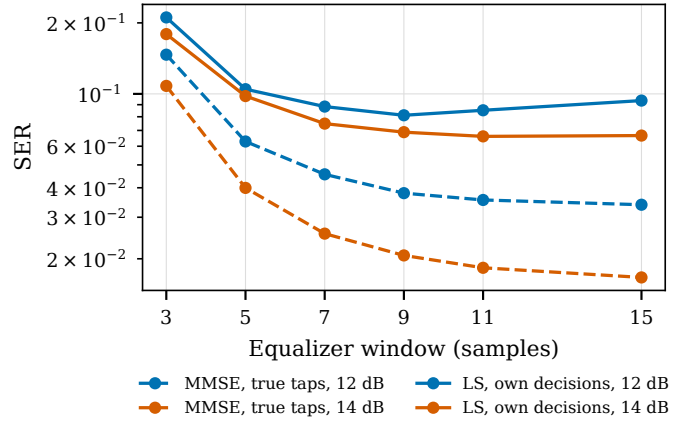


Fig. 16. QPSK linear equalizers vs. window length: MMSE weights from the true taps (dashed) and LS weights re-fitted on the equalizer's own decisions (solid).

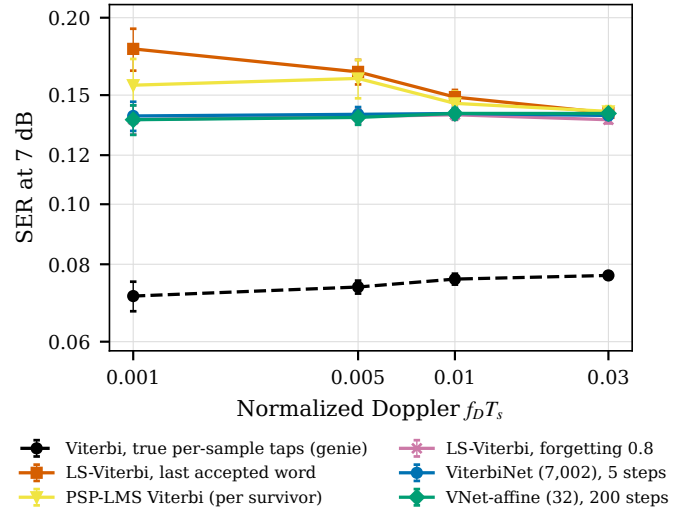


Fig. 17. BPSK SER at 7 dB vs. normalized Doppler (Rician $\kappa = 3$ Jakes fading within the word; 50 paired repetitions; bars: 95% confidence intervals).

to +0.19). Exponentially weighted LS is insensitive to its forgetting factor: any memory from 5 to 100 words ($\lambda = 0.8$ –0.99) ties VNet-affine within ± 0.002 at $f_D T_s \leq 0.01$ and beats it by 0.003–0.004 at 0.03 (95% CI ± 0.002). A memory of two words ($\lambda = 0.5$) already recovers most of the gap that LS on the last word leaves (+0.005 against +0.038 at $f_D T_s = 0.001$). Tuning therefore strengthens the conclusion: a classical estimator with memory matches the learned receivers over a wide range of its only parameter, and no setting of either tracker approaches the genie (0.069–0.078).

The complex QPSK channel is harsher (Fig. 19), because tap phases rotate. At $f_D T_s = 2 \times 10^{-4}$ (0.02 Doppler cycles per word) the trellis receivers are already 6–8 times worse than the genie at 10 dB and 13–15 times worse at 14 dB, where LS-Viterbi and the tied metric reach 0.099 and 0.101 against 7.7×10^{-3} . At $f_D T_s \geq 5 \times 10^{-4}$ every receiver that estimates one channel per word, learned or classical, trellis or equalizer, falls to 0.56–0.75, near chance (0.75), while the genie stays at 5 – 8×10^{-3} at 14 dB. The learned and

TABLE VI

PAIRED SER DIFFERENCES UNDER BPSK FAST FADING AT 7 dB (MEAN $\pm$ 95% CI OVER 50 REPETITIONS; POSITIVE: FIRST RECEIVER WORSE).

$f_D T_s$	0.001	0.005	0.01	0.03
LS (last word) – ViterbiNet $K=5$	$+0.039 \pm .011$	$+0.024 \pm .006$	$+0.009 \pm .003$	$+0.001 \pm .002$
RLS ($\lambda=0.8$) – ViterbiNet $K=5$	$-.002 \pm .001$	$-.001 \pm .001$	$-.001 \pm .001$	$-.002 \pm .002$
RLS ($\lambda=0.8$) – VNet-affine	$+0.000 \pm .001$	$+0.001 \pm .001$	$-.001 \pm .001$	$-.003 \pm .001$

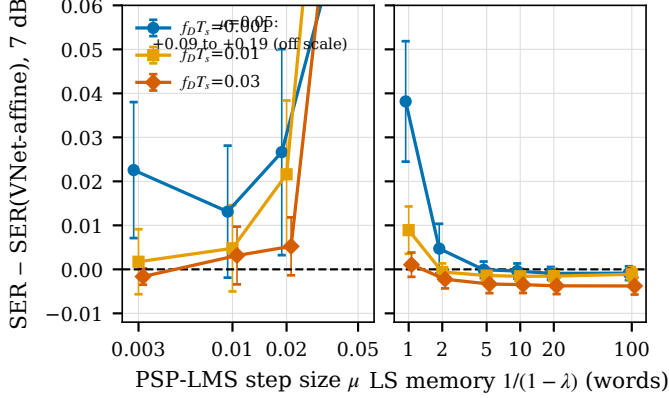


Fig. 18. Classical tracker settings under BPSK fast fading at 7 dB: SER minus that of VNet-affine on the same 20 repetitions (bars: 95% CI of the paired difference). (a) PSP-LMS step size. (b) LS forgetting factor, as effective memory in words; 1 is LS on the last accepted word.

classical receivers fail together and by the same amount, so the limitation is the per-word channel model shared by all of them, not the choice between learning and estimation. Handling such channels requires tracking within the word, which none of the learned receivers studied here provides.

VIII. DISCUSSION

When does learning help? In every setting studied, the branch metric that the learned receivers must represent is fixed by a linear channel and Gaussian noise: quadratic in the received sample, with L unknown taps. A classical receiver that estimates those taps from the same data reaches the bound up to the loss (3), and a learned metric can at best match it. Learning can add value where that model is wrong: nonlinear front ends, impulsive or non-Gaussian noise, or channels without a finite-tap description [4]. Our results argue that such claims should be tested against a matched-information classical receiver rather than against a Viterbi detector with corrupted CSI.

Where the learned receivers lose. The losses are small at low SNR and grow with SNR (Fig. 5). A metric trained by a few stochastic gradient steps on a noisy target has a residual error that stops mattering only when noise dominates. The same mechanism explains the overfitting of $K = 200$ (Fig. 7), the inefficiency of the attention metrics (Fig. 9), and the failure of free per-branch metrics for QPSK (Fig. 14): every added parameter must be estimated from the same few hundred labeled samples.

Design guidance. (i) Compare against a classical receiver that receives the same pilots and decisions. (ii) Tune the online adaptation budget; the right number of steps depends on model size and can differ by two orders of magnitude. (iii) Build the

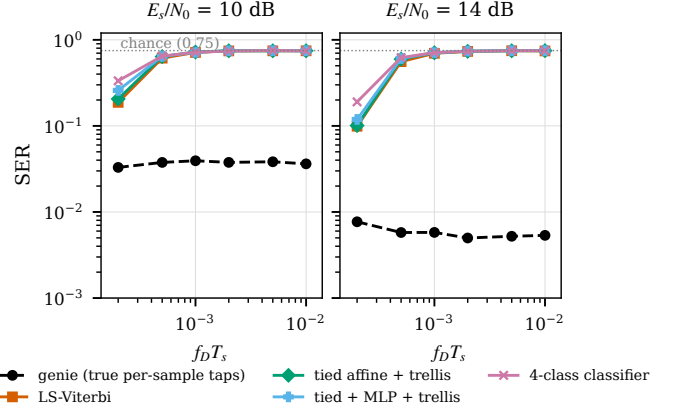


Fig. 19. QPSK SER vs. normalized Doppler with complex Jakes taps varying within the word (64 frames per point, identical draws for all receivers).

channel structure into the metric: for M -ary signalling, tie the M^L branch classes to L taps. (iv) Measure cost end to end against the symbol rate, and count operations as well as time; the adaptation loop, not the network forward pass, sets the cost of a learned receiver, while LS re-estimation costs less than the trellis itself. (v) Under fast fading, evaluate tracking within the word against PSP or exponentially weighted estimators before attributing gains to learning. (vi) Report results where they are statistically supported; above about 14 dB in our setting the learned receivers' SER rests on a handful of errors.

Limitations. Our BPSK results use one COST 2100 tap trace, a linear Gaussian channel, and our own implementation of ViterbiNet; the QPSK and fast-fading channels are synthetic. The meta-learning variants of [9], [10] change how the learned receivers adapt and should be compared against the same baselines. Latency was measured in an interpreted implementation on one CPU thread; absolute numbers and the share of the trellis will change on other platforms, whereas the operation counts of Fig. 3 do not.

IX. CONCLUSION

Given the same pilots and decisions, a least-squares Viterbi receiver matched or outperformed every learned trellis detector we tested from 0 to 14 dB, stayed within 3–30% of the perfect-CSI bound as a simple estimation-loss argument predicts, and needed a single least-squares solve per accepted word to adapt. ViterbiNet's gains are gains over a receiver denied the chance to estimate its channel. Within the learned family, fewer adaptation steps, fewer parameters, and structure tied to the channel model all helped, and attention-based metrics did not. The open problem these experiments expose is tracking within a word under fast fading, where every receiver without CSI, learned or classical, remains far from the bound.

DATA AVAILABILITY AND AI-USE STATEMENT

Simulation code, raw results, and the scripts that generate every figure and table (paper/make_figures.py) are available at <https://github.com/Gilzuk/viterbitransformed>. An AI coding assistant (Claude, Anthropic) was used to write simulation, analysis, and plotting code and to draft the manuscript; the author checked all code, results, and text.

REFERENCES

- [1] T. O'Shea and J. Hoydis, "An introduction to deep learning for the physical layer," *IEEE Trans. Cogn. Commun. Netw.*, vol. 3, no. 4, pp. 563–575, Dec. 2017.
- [2] H. Ye, G. Y. Li, and B.-H. Juang, "Power of deep learning for channel estimation and signal detection in OFDM systems," *IEEE Wireless Commun. Lett.*, vol. 7, no. 1, pp. 114–117, Feb. 2018.
- [3] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, 2019.
- [4] N. Farsad and A. Goldsmith, "Neural network detection of data sequences in communication systems," *IEEE Trans. Signal Process.*, vol. 66, no. 21, pp. 5663–5678, Nov. 2018.
- [5] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.
- [6] N. Shlezinger, N. Farsad, Y. C. Eldar, and A. J. Goldsmith, "ViterbiNet: A deep learning based Viterbi algorithm for symbol detection," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3319–3331, May 2020.
- [7] G. D. Forney, Jr., "The Viterbi algorithm," *Proc. IEEE*, vol. 61, no. 3, pp. 268–278, Mar. 1973.
- [8] —, "Maximum-likelihood sequence estimation of digital sequences in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 18, no. 3, pp. 363–378, May 1972.
- [9] T. Raviv, S. Park, N. Shlezinger, O. Simeone, Y. C. Eldar, and J. Kang, "Meta-ViterbiNet: Online meta-learned Viterbi equalization for non-stationary channels," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2021.
- [10] T. Raviv, S. Park, O. Simeone, Y. C. Eldar, and N. Shlezinger, "Online meta-learning for hybrid model-based deep receivers," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6415–6431, Oct. 2023.
- [11] F. R. Magee and J. G. Proakis, "Adaptive maximum-likelihood sequence estimation for digital signaling in the presence of intersymbol interference," *IEEE Trans. Inf. Theory*, vol. 19, no. 1, pp. 120–124, Jan. 1973.
- [12] R. Raheli, A. Polydoros, and C.-K. Tzou, "Per-survivor processing: A general approach to MLSE in uncertain environments," *IEEE Trans. Commun.*, vol. 43, no. 2/3/4, pp. 354–364, 1995.
- [13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 5998–6008.
- [14] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
- [15] S. Haykin, *Adaptive Filter Theory*, 5th ed. Upper Saddle River, NJ, USA: Pearson, 2014.
- [16] I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *J. Soc. Ind. Appl. Math.*, vol. 8, no. 2, pp. 300–304, 1960.
- [17] L. Liu, C. Oestges, J. Poutanen, K. Haneda, P. Vainikainen, F. Quitin *et al.*, "The COST 2100 MIMO channel model," *IEEE Wireless Commun.*, vol. 19, no. 6, pp. 92–99, Dec. 2012.
- [18] R. H. Clarke, "A statistical theory of mobile-radio reception," *Bell Syst. Tech. J.*, vol. 47, no. 6, pp. 957–1000, 1968.
- [19] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.